

LEO ZHOU

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Education

Rice University | Bachelor of Science in Computer Science | **GPA: 3.92/4.00** **Expected Graduation, May 2028**

- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Discrete Math, Linear Algebra

Technical Skills

Languages: Python, C/C++, Go, JavaScript/TypeScript, Swift, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, Flask, React, Node.js, SwiftUI, PyTorch, ONNX, REST APIs, WebSockets

Tools & Cloud: AWS (ECS, SQS, RDS), Docker, Terraform, PostgreSQL, Redis, SQLite, Git, GitHub Actions

Experience

Software Engineer Intern | **CanSemi**

June 2026 – August 2026

- Engineered an event-driven parser in Python extracting test measurements from 500+ binary test equipment (ATE) logs
- Designed a normalized PostgreSQL schema and batch ETL pipeline loading 40 wafer lots (~12,000 records) in <2 min
- Developed a REST API in Python and FastAPI with 8 endpoints, cutting report turnaround ~75% for 6 engineers
- Built an interactive data dashboard in React with hover and drill-down, replacing a manual Excel reporting process

Software Engineer | **Inkstone Technologies**

April 2026 – July 2026

- Built a native iOS reading application in Swift and SwiftUI that adapts content to each user's known vocabulary
- Integrated Apple Vision OCR into an on-device scanning pipeline, overlaying pronunciation on unfamiliar words only
- Wrote a Python pipeline to parse, normalize, and frequency-rank a 113,000-entry dictionary into an indexed SQLite DB
- Designed a local-first SQLite layer on the raw C API, driving offline spaced-repetition review and a WidgetKit widget

Embedded Software Engineer | **Rice Wind Energy**

October 2025 – April 2026

- Programmed ESP32 microcontroller firmware in C/C++ to monitor turbine speed, power output, and blade pitch
- Implemented a six-state safety state machine with fault detection and recovery, plus a peak power tracking algorithm

Projects

BlunderNet — Self-Training Chess Engine 🌀 | *Python, PyTorch, C++, pybind11, GitHub Actions*

- Trained an AlphaZero-style chess engine from scratch in PyTorch, a residual CNN learning purely from its own self-play
- Accelerated Monte Carlo Tree Search 1.8× by rewriting it in C++ and bridging back to Python with pybind11
- Automated retraining on GitHub Actions CI/CD, tracking strength against Stockfish and a 1,200-puzzle tactics suite

BlunderNet Arena — Cloud Serving Platform 🌀 | *Go, AWS, Terraform, Docker, Redis, React*

- Built a site where anyone can play the BlunderNet engine, on a stateless Go API fleet behind an AWS load balancer
- Scaled inference onto autoscaling ONNX workers behind AWS SQS, streaming live moves over WebSockets and Redis
- Provisioned the AWS environment (VPC, ECS, RDS, SQS) in Terraform, load-tested with k6 at p99 under 30 ms

SeatLive — Concurrency-Safe Booking System 🌀 | *Python, FastAPI, PostgreSQL, Redis, Docker*

- Built a FastAPI booking backend preventing double-booked seats with atomic transactions and idempotency keys
- Validated with CI tests firing 50 concurrent requests at one seat and Locust load tests at 4,459 requests, zero failures

Leadership & Athletics

- **New Zealand Junior National Fencing Team** — Captain, Coach, Referee; 30+ titles, Junior World Championships
- **Rice Fencing Club** — Manager · **Physics Pathfinders** — President; student-led charity, 600+ students, 22 schools